3(a)(i)	a gas that obeys $pV \propto T$	**M1**
	where p = pressure, V = volume, T = thermodynamic temperature	**A1**
3(a)(ii)	T = (273 + 17) K	**C1**
	$n = pV / RT$ $= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273 + 17)]$ = 12 mol	**A1**
3(b)(i)	work done $= p\Delta V$	**C1**
	$= 1.2 \times 10^5 \times (0.24 - 0.08)$ = 19200 J (= 19.2 kJ)	**A1**
3(b)(ii)	AB work done correct (19.2)	**A1**
	BC work done correct (0)	**A1**
	CA increase in internal energy correct (0) **and** CA thermal energy correct (31.6)	**A1**
	AB increase in internal energy calculated correctly from work done – 48.0	**A1**
	BC increase in internal energy correctly calculated so the final column adds up to zero **and** BC thermal energy same as increase in internal energy (Fully correct table: <table><tr><td>AB</td><td>19.2</td><td>–48.0</td><td>–28.8</td></tr><tr><td>BC</td><td>0</td><td>28.8</td><td>28.8</td></tr><tr><td>CA</td><td>–31.6</td><td>31.6</td><td>0</td></tr></table>)	**A1**